Subham Bhattacharya

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Education

Iowa State University — **Aug 2021 – Dec 2025**
Bachelor of Science in Computer Science, Minor in Data Science — *Ames, IA*

Relevant Coursework: Data Structures, Algorithms, Database Systems, AI, Object-Oriented Programming, Web Development, Mobile App Development

Technical Summary: Python, Java, C++, Android (Kotlin/Jetpack Compose), MERN stack (MongoDB, Express, React, Node)

Internship Experience

Qualcomm — **May 2024 – Aug 2024**
Software Engineering Intern — *San Diego, CA*

- Engineered a kernel-level CPU latency benchmarking library in C/C++ using Windows Driver Framework (WDF), integrating CPU clock time functionality into company-wide performance testing codebase to measure thread execution delays with nanosecond precision, **improving observability by 20%** for driver development teams
- Implemented IOCTL-based communication layer to enable seamless user-space to kernel-space interaction, allowing developers to trigger precise performance measurements and retrieve real-time latency metrics via command-line interface.
- Designed priority-based kernel thread scheduling with configurable core affinity, enabling accurate performance profiling across multi-core Windows systems for driver optimization workflows.

Relevant Experience

Codepath — **Aug 2025 – current**
Tech Fellow - TIP 101 (Technical Interview Prep) — *Remote*

- **Mentored 6+ students** per session in advanced data structures and algorithms, delivering personalized coaching on problem-solving patterns, complexity analysis, and optimization techniques across topics including backtracking, DFS, Binary Search, trees, and Object Oriented fundamentals to prepare students for technical interviews at top-tier companies.
- Collaborate with CodePath leadership team to refine curriculum materials and provide feedback on course content, contributing to **improved learning outcomes for 100+ students** across multiple cohorts.

Google Developers Student Club — **Oct 2022 – May 2025**
Tech Lead - Android — *Ames, IA*

- Architected and delivered a comprehensive Android development curriculum from foundational concepts to production-ready applications, **upskilling 30+ students** with zero mobile development experience to build fully functional apps using Jetpack Compose, Kotlin, and MVVM architecture.
- Designed hands-on technical workshops covering UI component development with Compose declarative framework, state management, REST API integration, Room database implementation, and dependency injection with Hilt/Dagger, resulting in **30+** students completing end-to-end projects.

Projects

FindRight - Service Matching Application | *Java, Android Studio, XML, GitLab CI/CD* — **Aug 2023 - Dec 2023**

- Architected and developed the front-end for a two-sided marketplace Android application connecting blue-collar service providers (plumbers, electricians, contractors) with clients seeking on-demand services, implementing role-based user flows and real-time service discovery features.
- Collaborated with 3-person backend team in Agile sprints to design and implement RESTful API integration, user authentication system, and scalable database schema supporting **50+** test users during beta phase.
- Established CI/CD pipeline using GitLab with automated unit and integration testing across **20+** core user flows, reducing bug detection time by **40%** and ensuring consistent deployment quality for profile management, service booking, and payment features
- **Demo:** https://youtu.be/mgoWAoikVoQ?si=-WssHbGfyTVa3HMY

Cyflix | *MongoDB, Express.js, React.js, Node.js* — **June 2024 - July 2024**

- Developed a full-stack video streaming application using MERN stack with JWT-based authentication, integrating TMDB REST API to dynamically render **1000+** movie titles with hover previews and seamless video playback functionality.
- Engineered MongoDB-backed watchlist system with real-time UI updates using React hooks and Context API, implementing secure session management and role-based access control for personalized content recommendations.
- **Demo:** https://youtu.be/9oS3cz72fZQ

Terminal Based Pokemon game | *C, C++, ncurses, CLion* — **Aug 2022 - Dec 2022**

- Developed a feature-rich roguelike Pokemon game in C/C++ with procedurally generated maps, turn-based combat system, and NPC trainers, implementing object-oriented design patterns and efficient memory management for smooth terminal-based gameplay.
- Engineered intelligent NPC pathfinding using Dijkstra's algorithm with heap-based priority queues, enabling dynamic trainer movement and realistic battle encounters across a persistent game world with multiple biomes and spawn mechanics.
- Optimized performance through manual memory management and pointer arithmetic, handling complex game state persistence, collision detection, and real-time UI rendering with ncurses library for responsive terminal graphics.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, Kotlin
Developer Tools: Git, VS Code, WinDbg, IntelliJ, Google Cloud Platform, Android Studio
Technologies/Frameworks: HTML/CSS, React, WDF (Windows Driver Framework), Linux, GitHub, GitLab, JUnit